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Module 1 Discussion
Module Assessment
Quiz: Graded Assessment - Integer Foundation
Submitted

Graded Assessment - Integer Foundation

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1. Which of the following pairs of numbers are relatively prime? **1 / 1 point**

- 796,982
- (selected) 819,990
- 91, 343
- 527,612

Correct
819 = 7 7 13, 990 = 2 3 3 5 11 2

2. What is GCD(930,992)? **1 / 1 point**

- 16
- (selected) 62
- 2
- 31

Correct
930 = 2 3 5 31, 992 = 2 2 2 2 3 1

3. Which of the following are congruent (mod 13)? **1 / 1 point**

- 3, 113
- -4, 4
- (selected) -17, 9
- 21001, 21013

Correct
The difference is 26, which is divisible by 13.

4. If x and -x to be congruent (mod N), what do you know about N? **1 / 1 point**

- N must be odd.
- N must be even.
- x must divide N
- (selected) x is a multiple of half the modulus.

Correct
N must divide (x - -x) = 2x.

5. What is sufficient for the multiplicative inverse of x (mod N) to exist? **1 / 1 point**

- x must not be 0.
- N must be prime.
- (selected) x must be relatively prime to N.
- x must be prime.

Correct
This is the only requirement.

6. The number of iterations required by the Euclidean Algorithm is **1 / 1 point**

- O(log(n)) (where n is the larger number).
- O(n) (where n is the smaller number).
- (selected) O(log(n)) (where n is the smaller number).

Correct
The first iteration reduces the problem to one the size of the smaller number.

7. What is the multiplicative inverse of 16 (mod 47) **1 / 1 point**

- 16
- (selected) 3
- -3
- 5

Correct
That this is true is clear since 3*16 = 48.

8. What is the multiplicative inverse of 219 (mod 220)? **1 / 1 point**

- 221
- It doesn't exist.
- 0
- (selected) -1

Correct
219 is congruent to -1 (mod 220) and -1*-1 = 1